The Complex Almost-Injectivity Threshold Is Exactly $2d$

A degree obstruction for every critical rank-one frame

Automated proof attempt for arXiv:1403.1458
`agent_lane_03` model: GPT5.6

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Status: full result, pending human verification. For every standard complex measurement system with $N \le 2d-1$, almost every signal has a nontrivial intensity twin. Combining this universal necessity with the known generic sufficiency at $N \ge 2d$ proves Conjecture 2 of the source paper.

Source question

Mixon defines almost injectivity as uniqueness modulo global phase on an open dense set and asks for a phase transition at $N = 2M$. The excerpt is from printed pages 22–23, Section 3.2 of the source paper.

Source p. 22

Take an almost injective intensity measurement mapping $\mathcal{A}$ and restrict it to an open set S of x's for which $\mathcal{A}^{-1}(\mathcal{A}(x)) = \{\omega x : |\omega| = 1\}$. Note that $\mathcal{A}$ is injective over S by assumption. Considering $(\mathbb{C}^M \setminus \{0\})/\mathbb{T}$ is a smooth manifold of real dimension $P = 2M-1$, we can intersect S with a patch to get an open set U, and consider the Jacobian of $\mathcal{A}$ in the patch's local coordinates. By the contrapositive of Theorem 16, we have that $N \ge \operatorname{rank}(J(x)) \ge P = 2M-1$ for some $x \in U$. This may lead one to believe that $\operatorname{AlmInj}[\Phi, \mathbb{C}^{M\times N}]$ exhibits a phase transition at $N = 2M-1$, but there is evidence to suggest that is this off by 1 due to algebraic properties of intensity measurements:

Conjecture 2. $\operatorname{AlmInj}[\Phi, \mathbb{C}^{M\times N}]$ exhibits a phase transition at $N = 2M$.

Source p. 23

To be clear, part (b) of this conjecture was proved by Balan, Casazza and Edidin [5], whereas a sketch of the proof of part (a) is provided in [31]. However, the latter sketch leaves much to be desired – while the argument is believable in principle, it is unclear whether their use of real algebraic geometry is sufficiently rigorous. For explicit minimal constructions in this case (assuming the conjecture is true), see [30, 31].

Result

For $\Phi = (\varphi_j)_{j=1}^N \subset \mathbb{C}^d$, write

$$\mathcal{A}_\Phi(x) = \big(|\langle x, \varphi_j\rangle|^2\big)_{j=1}^N, \qquad x \sim y \iff y = \omega x \quad (\omega \in \mathbb{T}).$$

Theorem 1 (Universal failure at and below the critical count)**.** *Let $d \geq 2$ and $N \leq 2d-1$. For every $\Phi \subset \mathbb{C}^d$, there is an open dense conull set $G \subset \mathbb{C}^d$ such that every $x \in G$ has a $y \not\sim x$ with $\mathcal{A}_\Phi(y) = \mathcal{A}_\Phi(x)$. Consequently $\mathcal{A}_\Phi$ is neither almost injective in the source's open-dense sense nor phase retrievable almost everywhere.*

This is stronger than merely producing one bad signal: at the proposed lower threshold, ambiguity is the generic and almost-everywhere behavior for *every* frame.

Proof intuition

Normalize a spanning frame to a Parseval frame. Its analysis map identifies signals with a d-plane $L \subset \mathbb{C}^{2d-1}$, and the measurements become the coordinate magnitudes of points of L. Fix a point $z \in L$. Changing its coordinate phases traces a torus of real dimension $2d-2$. Membership back in L imposes $d-1$ complex equations, also $2d-2$ real equations. Thus the critical ambiguity problem is the zero set of a square map

$$\mathbb{T}^{2d-2} \longrightarrow \mathbb{C}^{d-1} \cong \mathbb{R}^{2d-2}.$$

The original point is always a zero. If that zero is nondegenerate, it cannot be the only one: a map from a closed manifold into Euclidean space has degree zero, while one regular preimage would have mod-2 degree one. If the zero is not generically nondegenerate, the normalized intensity map has deficient maximal rank, and the constant-rank theorem gives positive-dimensional fibers. Either way, generic uniqueness is impossible.

Proof

1. Parseval and projective reductions

If the measurement vectors do not span $\mathbb{C}^d$, a nonzero common-kernel vector may be added to every signal outside a proper subspace without changing any measurement. Hence assume they span. Let $S = \sum_j \varphi_j\varphi_j^*$, put $g_j = S^{-1/2}\varphi_j$, and replace x by $S^{1/2}x$. Then

$$|\langle S^{1/2}x, g_j\rangle| = |\langle x, \varphi_j\rangle|, \qquad \sum_j g_j g_j^* = I_d.$$

Thus no fiber changes under an invertible signal coordinate change. The analysis operator

$$V : \mathbb{C}^d \to \mathbb{C}^N, \qquad Vu = (\langle u, g_j\rangle)_j$$

is an isometry. Set $L = \operatorname{ran} V$.

On complex projective space define the normalized coordinate-intensity map

$$q : \mathbb{P}(L) \longrightarrow \Delta^{N-1}, \qquad q([z]) = \frac{(|z_1|^2, \ldots, |z_N|^2)}{\|z\|^2},$$

where $\Delta^{N-1} = \{a \in \mathbb{R}^N : a_j \geq 0,\ \sum a_j = 1\}$. It is real analytic, and $\dim_{\mathbb{R}} \mathbb{P}(L) = m := 2d-2$.

Let r be the maximum rank of dq. Its maximal-rank locus is open, dense, and conull: the rank-drop locus is the zero set of the nonzero real-analytic section $\bigwedge^r dq$ (with the constant-map case immediate). If $r < m$, then near every maximal-rank point the constant-rank theorem gives a local fiber of dimension $m - r > 0$. This proves the theorem whenever $N - 1 < m$, hence whenever $N < 2d-1$, and also handles the singular alternative at $N = 2d-1$.

It remains to treat $N = 2d-1$ and $r = m = N-1$.

2. The square phase map

The full-rank locus of q is open dense and conull. No coordinate functional can vanish identically on L, since then q would land in a face of dimension $N-2<m$. Hence the projective points whose coordinates are all nonzero also form an open dense conull set. Fix a unit representative z in the intersection of these two sets.

Choose an isometry $K:\mathbb{C}^{d-1}\to\mathbb{C}^N$ with $\operatorname{ran} K=L^\perp$. Fixing the first coordinate phase removes global phase, and define

$$\Psi_z:\mathbb{T}^{N-1}\longrightarrow\mathbb{C}^{d-1},\qquad \Psi_z(\lambda_2,\dots,\lambda_N)=K^*\operatorname{diag}(1,\lambda_2,\dots,\lambda_N)z.$$

The identity $\mathbf{1}$ is a zero because $z\in L=\ker K^*$. Any other zero yields another vector of L with exactly the same coordinate magnitudes.

Lemma 2 (Transversality equivalence)**.** *The differential $dq_{[z]}$ is nonsingular if and only if the zero $\mathbf{1}$ of Ψ_z is nondegenerate.*

Proof. Represent a tangent class at $[z]$ by $\delta\in L$ orthogonal to z. Then

$$(dq_{[z]}\delta)_j=2\operatorname{Re}(\overline{z}_j\delta_j).$$

Since every $z_j\neq 0$, a kernel vector has the form $\delta_j=ih_jz_j$ with $h_j\in\mathbb{R}$. Subtracting the global infinitesimal phase ih_1z gives the same projective tangent class and sets $h_1=0$. The remaining condition $\delta\in L$ is exactly

$$K^*i\operatorname{diag}(0,h_2,\dots,h_N)z=0,$$

which is the kernel equation for $D\Psi_z(\mathbf{1})$. Conversely, a vector in this latter kernel gives a zero tangent class for dq; because $h_1=0$ and every coordinate of z is nonzero, the class is trivial only when all $h_j=0$. Both real vector spaces have dimension $N-1=2d-2$, so injectivity is equivalent to nonsingularity. □

3. A Euclidean-target degree obstruction

Lemma 3 (One regular zero is impossible)**.** *Let $n>0$ and let $F:\mathbb{T}^n\to\mathbb{R}^n$ be smooth. A nondegenerate zero of F cannot be its unique zero.*

Proof. Suppose x_0 were the unique zero and $DF(x_0)$ were invertible. Compose F with the one-point compactification $\mathbb{R}^n\hookrightarrow S^n$. The resulting map misses infinity and is therefore null-homotopic, so its mod-2 degree is zero. But zero is a regular value with the single preimage x_0, so its mod-2 degree is one, a contradiction. □

At our chosen $[z]$, Lemma 2 makes $\mathbf{1}$ a nondegenerate zero of Ψ_z. Lemma 3 supplies a second zero λ. Put $w=\operatorname{diag}(1,\lambda_2,\dots,\lambda_N)z$. Then $K^*w=0$, so $w\in L$, and $|w_j|=|z_j|$ for all j. If $w=\omega z$, the first nonzero coordinate forces $\omega=1$, and the remaining nonzero coordinates force $\lambda=\mathbf{1}$, contrary to the choice of the second zero. Thus $[w]\neq[z]$.

We have proved ambiguity throughout an open dense conull subset of $\mathbb{P}(L)$. Lifting projective points through radius and global phase, then undoing the invertible Parseval coordinate change, gives the asserted open dense conull set in $\mathbb{C}^d$. This proves Theorem 1. □

The phase transition

Corollary 4 (Conjecture 2 of arXiv:1403.1458)**.** *Standard complex almost injectivity exhibits a phase transition at $N=2d$.*

Proof. Theorem 1 proves part (a): no system below $2d$ is almost injective. Balan–Casazza–Edidin prove that for every $N\geq 2d$, a generic complex frame has a dense open set of uniquely recoverable rays [2, Theorem 3.4]. This is the part (b) already cited as known in the source paper [1, pp. 22–23]. □

Checks, scope, and literature status

The proof is topological and does not depend on computation. The included script `code/verify_torus_obstruction.py` deterministically builds random critical Parseval frames in dimensions $2, 3, 4$, checks both Jacobian ranks in Lemma 2, and numerically finds multiple phase zeros. It is a stress test only.

The rank-one hypothesis is essential. Huang–Rong–Wang–Xu construct $2d-1$ *general* Hermitian quadratic measurements with PR-ae, while explicitly recording the exceptional $2d-1$ standard rank-one problem as open and proving only generic failure [3, Section 5, Theorem 5.3]. Their construction does not have the plane–coordinate-torus structure used above.

For novelty triage, exact-phrase and citation searches were run through 11 August 2026 for "$2d-1$", "PR-ae", "standard phase retrieval", and "almost injective", including the citing literature of [3]. No later resolution of the exceptional rank-one case was found. This bounded search supports human review; it is not an exhaustive priority claim.

References

[1] D. G. Mixon, *Phase Transitions in Phase Retrieval*, arXiv:1403.1458 (2014), especially Section 3.2 and Conjecture 2. Included as `source_paper.pdf`.

[2] R. Balan, P. Casazza, and D. Edidin, *On signal reconstruction without phase*, Appl. Comput. Harmon. Anal. 20 (2006), 345–356; arXiv:math/0412411. Included under `references/`.

[3] M. Huang, Y. Rong, Y. Wang, and Z. Xu, *Almost Everywhere Generalized Phase Retrieval*, Appl. Comput. Harmon. Anal. 50 (2021), 16–33; arXiv:1909.08874. Included under `references/`.